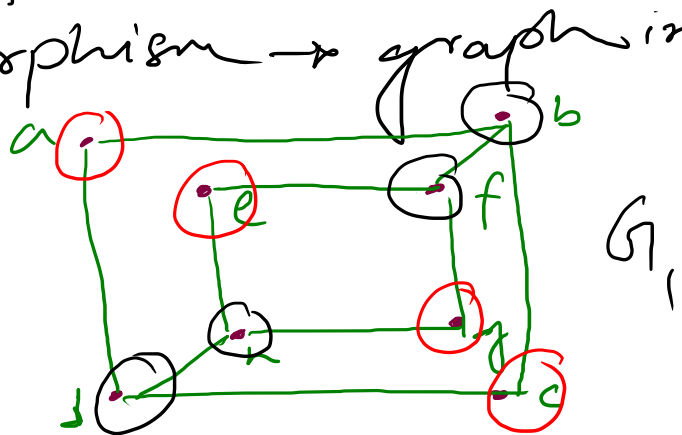


Isomorphism

- ① Same no. of elements despite different structure.
- ② A property preserved by isomorphism $\rightarrow$ graph invariant



* Conditions

① Same no. of vertices.

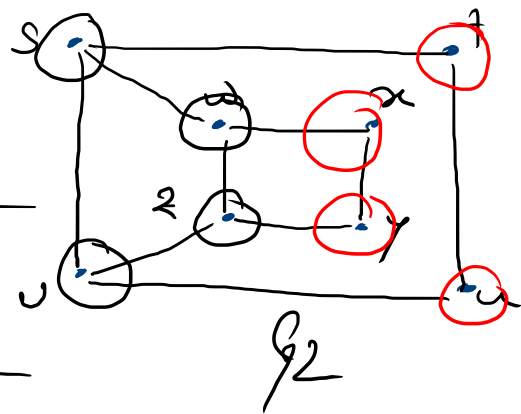
② " " " edges

③ " " " vertices with same degree

④

isomorphism function.

	G_1	G_2
# V	8	8
# E	10	10
# V	$V^3 \rightarrow 6$	$V^3 \rightarrow 4$
with same deg	$V^2 \rightarrow 4$	$V^2 \rightarrow 4$



$$A_{G_1} = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g & h \end{matrix} \\ \begin{matrix} \rightarrow a \\ \rightarrow b \\ c \\ d \\ e \\ f \\ \rightarrow g \\ \rightarrow h \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

not preserved

$$A_{G_2} = \begin{matrix} & \begin{matrix} s & t & u & v & w & x & y & z \end{matrix} \\ \begin{matrix} s \\ t \\ a \rightarrow u \\ v \\ w \\ x \\ g \rightarrow y \\ h \rightarrow z \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

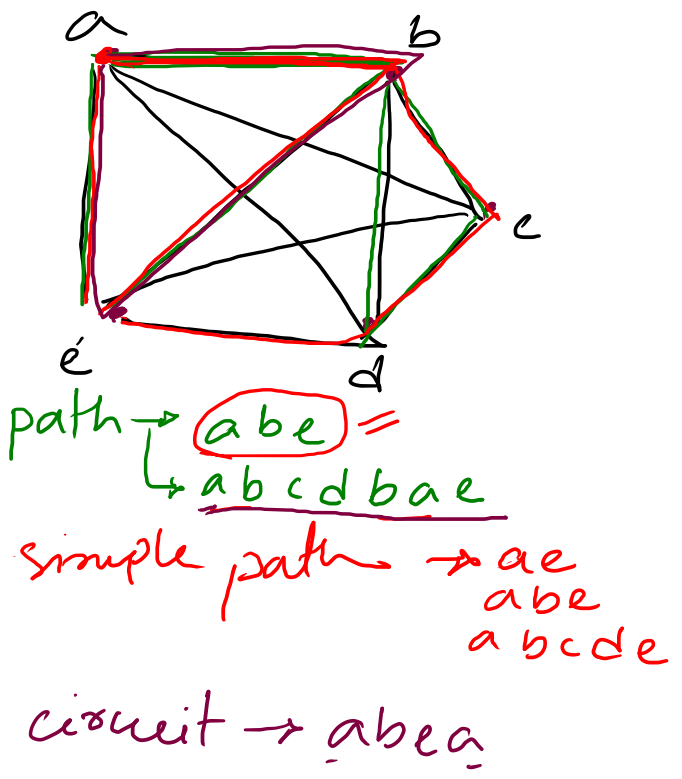
G_1, G_2 are not isomorphic.

10.4. Connectivity

path $v_1 \dots v_2$

simple path $\rightarrow$ with no repeating edges.

circuits $v_1 \dots v_2$

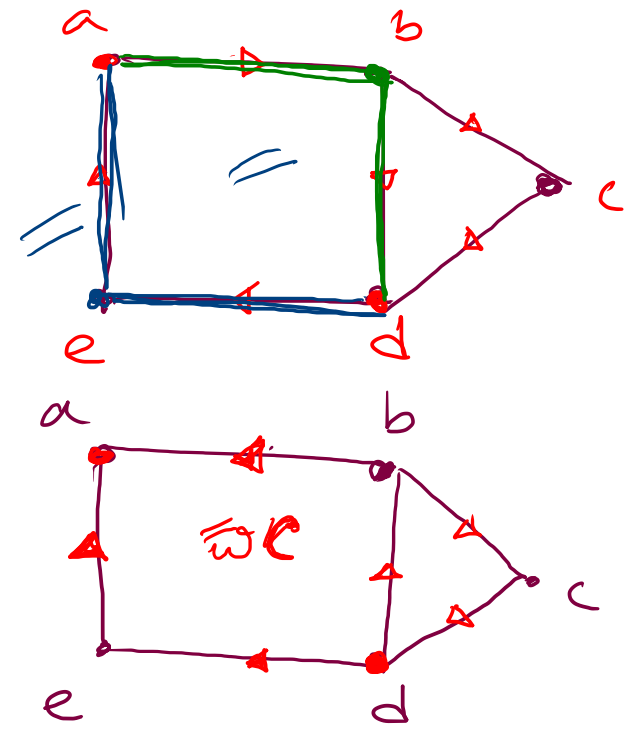


Connected Graph $\rightarrow$ uG

DG

$\rightarrow$ Strongly Connected
 $\hookrightarrow$ $\{ \text{path}(a, b)$
 $\text{path}(b, a) \}$

$\rightarrow$ Weakly connected.
 $\hookrightarrow$



10.5 Euler + Hamiltonian path

① Euler

- path → a simple path with no repeating edges.
- ckt → a simple ckt " " " "
- every vertex must have even degree.
- every edge has to be present exactly once. while vertices can be repeated.

② Hamiltonian.

→ passes through every vertex exactly once.

→ a graph with vertex of degree 1 will not have HC.

